The type-2 obstruction to a direct UMD extension of sharp Besov–Orlicz stochastic regularity

Candidate answer to the open question in arXiv:1901.01018

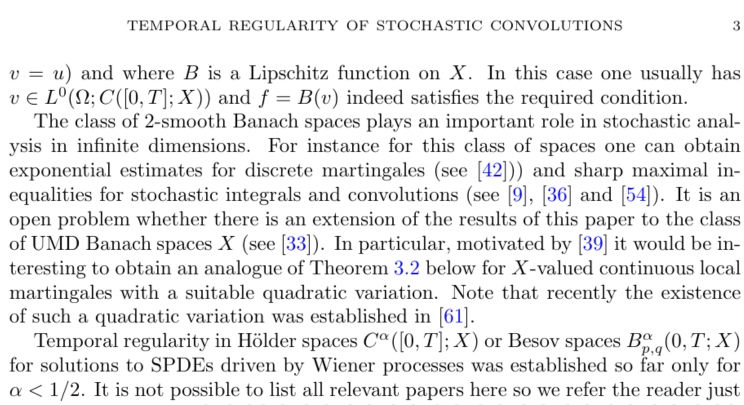
11 August 2026

Result. A direct extension to all UMD spaces, retaining the paper's $L^\infty_t\gamma(H,X)$ data class, is false. The UMD space ℓ^r, $1<r<2$, gives an explicit counterexample. More precisely, among separable UMD spaces the direct endpoint estimate holds exactly for spaces of type 2; these are exactly the UMD spaces that admit an equivalent 2-smooth norm. This does not settle a differently formulated result based on stronger, direct control of the full quadratic variation.

1 Source question and scope

Ondreját and Veraar prove sharp temporal regularity in $B^{1/2}_{\Phi_2,\infty}(0,T;X)$ for stochastic integrals and analytic stochastic convolutions when X is 2-smooth [1]. On page 3 they ask:

> Is there an extension of the results of the paper to UMD Banach spaces X? In particular, can one obtain an analogue of their stochastic-integral theorem for X-valued continuous local martingales with a suitable quadratic variation?



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$v=u$) and where B is a Lipschitz function on X. In this case one usually has $v\in L^0(\Omega;C([0,T];X))$ and $f=B(v)$ indeed satisfies the required condition.

The class of 2-smooth Banach spaces plays an important role in stochastic analysis in infinite dimensions. For instance for this class of spaces one can obtain exponential estimates for discrete martingales (see [42])) and sharp maximal inequalities for stochastic integrals and convolutions (see [9], [36] and [54]). It is an open problem whether there is an extension of the results of this paper to the class of UMD Banach spaces X (see [33]). In particular, motivated by [39] it would be interesting to obtain an analogue of Theorem 3.2 below for X-valued continuous local martingales with a suitable quadratic variation. Note that recently the existence of such a quadratic variation was established in [61].

Temporal regularity in Hölder spaces $C^\alpha([0,T];X)$ or Besov spaces $B^\alpha_{p,q}(0,T;X)$ for solutions to SPDEs driven by Wiener processes was established so far only for $\alpha<1/2$. It is not possible to list all relevant papers here so we refer the reader just to some of them, e.g. [16], [8], [40], [32], [48], [49], [10], [18], [35], [34], [14], [58], [4],

Figure 1: The open question in the source paper, page 3.

The phrase "an extension" permits changed hypotheses. We therefore isolate the most literal and useful version: retain the source paper's pointwise data class $L^\infty(0,T;\gamma(H,X))$ and its endpoint

Besov–Orlicz estimate. The theorem below completely characterizes this direct version inside the UMD class. The broader quadratic-variation formulation is discussed separately.

Recall that X has (Gaussian) type 2 if a constant $T_2(X)$ exists such that, for all finite families $x_1, \dots, x_n \in X$,

$$\left(\mathbb{E}\left\|\sum_{k=1}^{n} \gamma_k x_k\right\|^2\right)^{1/2} \le T_2(X) \left(\sum_{k=1}^{n} \|x_k\|^2\right)^{1/2},$$

where the γ_k are independent standard Gaussians. Gaussian and Rademacher type 2 are equivalent up to universal constants.

2 Proof intuition

Two elementary reductions reveal the exact geometry. First set the generator to zero and drive the equation by one real Brownian motion. Put a different vector x_k on each of several disjoint time intervals. At the end, the integral is exactly a Gaussian sum $\sum \gamma_k x_k$. If all intervals occur before time $1/2$, the path is constant on the remaining half of the time axis, so its inhomogeneous Besov–Orlicz norm controls the norm of that Gaussian sum.

Thus the endpoint regularity estimate forces type 2. In the other direction, UMD plus type 2 gives martingale type 2 by applying the UMD randomization inequality and then type 2 pointwise. Pisier's renorming theorem turns martingale type 2 into an equivalent 2-smooth norm, exactly the setting of the source paper. The explicit failure in ℓ^r, $r < 2$, comes from choosing interval lengths whose sum is finite but whose $r/2$-powers have divergent sum.

3 The characterization

Let W be a standard real Brownian motion and, for a deterministic simple $f : (0,1) \to X$, write

$$M_f(t) = \int_0^t f(s)\, dW(s).$$

Consider the direct endpoint estimate

$$\mathbb{E}\|M_f\|_{B^{1/2}_{\Phi_2,\infty}(0,1;X)} \le C_X \|f\|_{L^\infty(0,1;X)}. \tag{D}$$

This is Theorem 5.1(iii) of the source with $A = 0$, $H = \mathbb{R}$, and $q = \infty$, restricted to deterministic simple integrands.

Theorem 1 (Direct-extension characterization)**.** *Let X be a separable UMD Banach space. The following are equivalent.*

1. *The direct endpoint estimate* (D) *holds.*
2. *X has type* 2*.*
3. *X has martingale type* 2*.*
4. *X admits an equivalent* 2*-smooth norm.*

*Under these conditions all the stochastic-integral and analytic stochastic-convolution conclusions of Theorems 3.2 and 5.1 of the source paper hold, with constants adjusted for the equivalent norm. Consequently, the paper's same-data results extend within the UMD class exactly to the type-*2 *UMD spaces.*

Proof. Assume (D), and fix $x_1, \ldots, x_n \in X$. Put $\sigma^2 = \sum_{k=1}^n \|x_k\|^2$ and discard zero vectors. Choose disjoint intervals $I_k \subset (0, 1/2)$ of lengths

$$a_k = \frac{\|x_k\|^2}{2\sigma^2},$$

and define $f = x_k/\sqrt{a_k}$ on I_k and $f = 0$ elsewhere. Then

$$\|f\|_{L^\infty(0,1;X)} = \sqrt{2}\,\sigma, \qquad M_f(t) = \sum_{k=1}^n \gamma_k x_k \quad (1/2 \le t \le 1),$$

for independent standard Gaussians γ_k.

The inhomogeneous Besov–Orlicz norm contains the Luxemburg L^{Φ_2} norm. If a function equals z on a set of measure $1/2$, then

$$\|z\mathbf{1}_E\|_{L^{\Phi_2}(0,1;X)} \ge \frac{\|z\|}{\sqrt{\log 3}},$$

because $\frac{1}{2}(\exp((\|z\|/\lambda)^2) - 1) \le 1$ forces $\lambda \ge \|z\|/\sqrt{\log 3}$. Hence (D) gives

$$\mathbb{E}\left\|\sum_{k=1}^n \gamma_k x_k\right\| \le \sqrt{2\log 3}\, C_X \left(\sum_{k=1}^n \|x_k\|^2\right)^{1/2}.$$

The Gaussian Kahane–Khintchine inequality upgrades the first moment on the left to the second moment. Thus X has type 2.

Now assume that X is UMD and has type 2. Let $(d_k)_{k=1}^n$ be an X-valued martingale difference sequence in L^2. If $\beta_{2,X}$ is a UMD constant, applying the UMD inequality to $(\varepsilon_k d_k)$ and the same deterministic signs once more gives, for every sign vector ε,

$$\mathbb{E}\left\|\sum_k d_k\right\|^2 \le \beta_{2,X}^2 \mathbb{E}\left\|\sum_k \varepsilon_k d_k\right\|^2.$$

Averaging in independent Rademacher signs and applying type 2 for each fixed sample point yields

$$\mathbb{E}\left\|\sum_k d_k\right\|^2 \le \beta_{2,X}^2 T_2(X)^2 \sum_k \mathbb{E}\|d_k\|^2.$$

Thus X has martingale type 2. Conversely, martingale type 2 implies type 2 by taking independent symmetric differences.

Pisier's renorming theorem states that martingale type 2 is equivalent to the existence of an equivalent 2-smooth norm [2]; this equivalence is also recalled explicitly in the source paper. This proves the equivalence of the four assertions.

Finally, equip X with the equivalent 2-smooth norm and apply Theorems 3.2 and 5.1 of [1]. Equivalent norms give equivalent $\gamma(H, X)$, Bochner–Orlicz, Besov–Orlicz, and probability norms, and do not change analyticity of the semigroup. The conclusions therefore transfer back to the original norm, with modified constants. $\square$

4 An explicit UMD counterexample

Theorem 2. *Fix $1 < r < 2$ and $X = \ell^r$, which is a separable UMD space. There are deterministic integrands $f_N \in L^\infty(0,1;X)$ with $\|f_N\|_{L^\infty} = 1$ such that*

$$\mathbb{E}\left\|\int_0^{\cdot} f_N(s)\,dW(s)\right\|_{B^{1/2}_{\Phi_2,\infty}(0,1;X)} \longrightarrow \infty.$$

There is also a deterministic $f \in L^\infty(0,1;X)$ of norm 1 for which the integral has no X-valued limit at time $1/2$.

Proof. Choose ρ with $1 < \rho \le 2/r$ and set

$$a_k = ck^{-\rho}, \qquad c = \left(2\sum_{j=1}^{\infty} j^{-\rho}\right)^{-1}.$$

Then $\sum_k a_k = 1/2$ but $\sum_k a_k^{r/2} = \infty$. Partition $(0,1/2)$ into consecutive intervals I_k of lengths a_k. With (e_k) the unit vector basis of ℓ^r, define

$$f_N(t) = \sum_{k=1}^{N} \mathbf{1}_{I_k}(t) e_k.$$

Clearly $\|f_N\|_{L^\infty} = 1$. For $t \in [1/2, 1]$ its stochastic integral is the constant random vector

$$Z_N = \sum_{k=1}^{N} \sqrt{a_k}\,\gamma_k e_k, \qquad \|Z_N\|_r = \left(\sum_{k=1}^{N} a_k^{r/2}|\gamma_k|^r\right)^{1/r}.$$

Put $b_k = a_k^{r/2}$, $S_N = \sum_{k\le N} b_k|\gamma_k|^r$, $m_r = \mathbb{E}|\gamma_1|^r$, and $v_r = \mathbb{E}|\gamma_1|^{2r}$. Then

$$\mathbb{E}S_N = m_r \sum_{k\le N} b_k, \qquad \mathbb{E}S_N^2 \le v_r \left(\sum_{k\le N} b_k\right)^2.$$

Paley–Zygmund therefore gives a constant $p_r > 0$, independent of N, for which

$$\mathbb{P}\left(S_N \ge \tfrac{1}{2} m_r \sum_{k\le N} b_k\right) \ge p_r.$$

Consequently

$$\mathbb{E}\|Z_N\|_r \ge p_r \left(\tfrac{1}{2} m_r \sum_{k\le N} a_k^{r/2}\right)^{1/r} \longrightarrow \infty.$$

The same Luxemburg-norm lower bound used above shows that the expected Besov–Orlicz norm is at least $(\log 3)^{-1/2}\mathbb{E}\|Z_N\|_r$, proving the first claim.

For the second claim take $f(t) = \sum_{k\ge 1} \mathbf{1}_{I_k}(t)e_k$. The formal terminal value has rth power of its norm equal to $\sum_k b_k|\gamma_k|^r$. The events $\{|\gamma_k| \ge 1\}$ are independent and have positive density almost surely; since b_k is decreasing and $\sum b_k = \infty$, the weighted subseries diverges almost surely. Hence the Gaussian series is not ℓ^r-valued. The partial integrals cannot converge in X as $t \uparrow 1/2$, even though $\|f(t)\|_r = 1$ almost everywhere. $\square$

5 Quadratic variation, limitations, and novelty

Yaroslavtsev's UMD Burkholder–Davis–Gundy theorem attaches to a UMD-valued martingale a covariance bilinear form and its Gaussian characteristic [3]. In the example above that characteristic is precisely the γ-norm of the coordinate series, and it diverges. Thus the example shows that pointwise control of $\|f(t)\|_{\gamma(H,X)}$ does not control the full covariance characteristic outside type 2. It does not contradict a future theorem that assumes the latter control directly, so the source's broader phrase "with a suitable quadratic variation" remains open in this packet.

A bounded search of the local run indexes and arXiv/web results used the exact paper title and combinations of "UMD", "Besov–Orlicz", "temporal regularity", "stochastic convolution", "continuous local martingale", and "quadratic variation". It found the source, the UMD quadratic-variation paper, and later applications of the 2-smooth result, but no exact statement of this counterexample or characterization. The ingredients are standard, so novelty confidence is moderate rather than definitive. Human review should focus on the intended meaning of "extension" and on the transfer through Pisier's equivalent norm.

References

[1] M. Ondreját and M. Veraar, *On temporal regularity of stochastic convolutions in 2-smooth Banach spaces*, Ann. Inst. Henri Poincaré Probab. Stat. **56** (2020), 1792–1808; arXiv:1901.01018.

[2] G. Pisier, *Martingales in Banach Spaces*, Cambridge Studies in Advanced Mathematics 155, Cambridge University Press, 2016.

[3] I. Yaroslavtsev, *Burkholder–Davis–Gundy inequalities in UMD Banach spaces*, Comm. Math. Phys. **379** (2020), 417–459; arXiv:1807.05573.